

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2022–23

Paper: CS-104 — Computer Organization & Architecture

Time: Three Hours

Maximum Marks: 70

*Instructions: Attempt any **five** questions including Question 1 which is compulsory.***Question 1.**

[5 + 5 = 10]

- (a) State True or False and justify briefly:
- (i) All high-level languages are compiled.
 - (ii) All low-level languages are compiled.
 - (iii) Storage is not a part of data processing.
 - (iv) Concept of jitter is related to audio-video data.
 - (v) Port address is used for source to destination delivery.
- (b) Briefly define the following:
- (i) Malware
 - (ii) Browser Hijacker
 - (iii) Macro Virus
 - (iv) FTP

Question 2.

[10 + 5 = 15]

- (a) What is data processing? What are its different types? Explain each of them briefly.
- (b) Explain the concept of stored program computers. Why is it considered better than hard-wired computers?

Question 3.

[10 + 5 = 15]

- (a) What do you understand by a web crawler? Explain the working of a web crawler. What type of pages cannot be accessed by a web crawler?
- (b) What is personalization of search results? State some of the parameters used for personalization of search results.

Question 4.

[10 + 5 = 15]

- (a) Explain the different types of network topologies with examples. Also give their advantages and disadvantages.
- (b) Explain the different components of a communication system.

Question 5.

[10 + 5 = 15]

- (a) What is the role of Internet Service Providers (ISPs) in the Internet? Explain their different types.
- (b) Differentiate between protocol and standards. What are the different types of standards used in communication? Name any one standard organization.

Question 6.

[10 + 5 = 15]

- (a) Differentiate between a client and a server. Explain the roles of an e-mail client and e-mail server. How does message transfer take place in an e-mail communication system? Explain.
- (b) Explain the applications of ICT in Education.

Question 7.

[10 + 5 = 15]

- (a) Explain the memory hierarchy of a computer system.
- (b) Differentiate between a CPU and a GPU. List some tasks where a GPU performs better than a CPU and explain why.